（5.5）式的严格推导结果

> Fixed-cutoff 端点极限、moving-cutoff 缺口与最强可证修正版定理

1. 精确问题 statement

目标是解释 printed page 1104 的 equation (5.5) 是否能从前文严格推出。核心对象如下。对 $t < t_0$
，

$$\rho_{X_0,t_0}(F,t) = \frac{1}{4\pi(t_0-t)} \exp\left(-\frac{|F-X_0|^2}{4(t_0-t)}\right),$$

并且对一个 fixed original-coordinate cutoff $\phi \in C_c^\infty(B_{2r}(X_0))$、$\phi = 1$ on $B_r(X_0)$,

$$\Psi(X_0,t_0;t) = \int_{\Sigma_t} (\cos\alpha)^{-p} \phi(F) \rho_{X_0,t_0}(F,t)\, d\mu_t.$$

Theorem 3.3 的 relevant estimate 是

$$\frac{d}{dt}\left(e^{c_1\sqrt{t_0-t}}\Psi(X_0,t_0;t)\right) \le -\text{nonnegative square terms} + c_2 e^{c_1\sqrt{t_0-t}}$$

其中关键点是 right side 有 one-sided positive error。Section 5 在 first singular point (X_0,T) 附近使用

$$F_\lambda(x,s) = \lambda(F(x,T+\lambda^{-2}s) - X_0), \qquad d\mu_s^\lambda = \lambda^2 d\mu_{T+\lambda^{-2}s},$$

并写 $w_\lambda = (\cos\alpha_\lambda)^{-p}$。printed equation (5.5) 的 literal endpoint quantity 是

$$G_\lambda(s) = e^{c_1\sqrt{T-(T+\lambda^{-2}s)}} \int_{\Sigma_s^\lambda} w_\lambda\, \phi_R(F_\lambda)\, (-s)^{-1} e^{-|F_\lambda|^2/[4(-s)]}\, d\mu_s^\lambda,$$

其中 ϕ_R is a compact cutoff fixed in scaled coordinates。要证明或诊断的是：对 fixed $-1 < s_1 < s_2 < 0$,

$$G_\lambda(s_2) - G_\lambda(s_1) \to 0 \qquad (\lambda \to \infty).$$

当前 continuation 要求的是 remaining moving-cutoff lemma 的 full proof；结论是：literal compact-scaled-cutoff lemma 在 full Section 5 compact first-singular-flow context 下尚未被证明，也没有 full-hypothesis counterexample；已证明的是 fixed-original-cutoff correction。

2. 主要数学进展

（status: **proven**）Fixed quantity 的 terminal limit 是严格成立的。设 $Q:[a,T)\to[0,\infty)$ locally absolutely continuous，并满足

$$Q'(t)\le c_2e^{c_1\sqrt{T-t}}$$

for almost every $t<T$。令

$$A(t)=\int_t^T c_2e^{c_1\sqrt{T-u}}\,du,\qquad M(t)=Q(t)+A(t).$$

则 $A(t)\to 0$ as $t\uparrow T$，且

$$M'(t)=Q'(t)-c_2e^{c_1\sqrt{T-t}}\le 0.$$

所以 M monotone decreasing 且 bounded below by 0，从而 $M(t)$ 有 finite one-sided limit。于是 $Q(t)=M(t)-A(t)$ 也有 finite one-sided limit。由此若 $t_\lambda(s)=T+\lambda^{-2}s$，则 fixed $s_1,s_2<0$ 给出

$$Q(t_\lambda(s_2))-Q(t_\lambda(s_1))\to 0.$$

这完全处理了 one-sided inequality 和 positive error term 的问题。

（status: **proven**）正确的 fixed-cutoff rescaled endpoint theorem 如下。固定一个 original-coordinate cutoff $\chi\in C_c^\infty(B_{2r}(X_0))$，$\chi=1$ on $B_r(X_0)$，定义

$$Q_\chi(t)=e^{c_1\sqrt{T-t}}\Psi_\chi(X_0,T;t).$$

在 scaled variables 中使用 expanding cutoff

$$\chi^\lambda(Y)=\chi(X_0+\lambda^{-1}Y).$$

则

$$G_\lambda^\chi(s)=e^{c_1\lambda^{-1}\sqrt{-s}}\int_{\Sigma_s^\lambda} w_\lambda\,\chi(X_0+\lambda^{-1}F_\lambda)(-s)^{-1}e^{-|F_\lambda|^2/[4(-s)]}\,d\mu_s^\lambda$$

满足 exact identity

$$G_\lambda^\chi(s)=4\pi Q_\chi(T+\lambda^{-2}s).$$

proof sketch：用

$$T-t=\lambda^{-2}(-s),\qquad F-X_0=\lambda^{-1}F_\lambda,\qquad d\mu_s^\lambda=\lambda^2d\mu_t.$$

Kähler angle 在 positive homothety 下 invariant，因为 numerator $\omega(u,v)$ 和 area denominator 都乘以 λ^2，所以 $w_\lambda=w$。于是 unnormalized scaled Gaussian factor 精确变成

$4\pi\rho_{X_0,T}$。因此 4π 只是 common normalization factor；若使用 normalized kernel $[4\pi(-s)]^{-1}$，这个因子消失。因为 $Q_\chi(t)$ has a common terminal limit，epsilon proof 直接给出

$$G^\chi_\lambda(s_2) - G^\chi_\lambda(s_1) \to 0.$$

（status: **proven**）printed cutoff 与 fixed cutoff 不是同一个对象。literal expression 中

$$\phi_R(F_\lambda) = \phi_R(\lambda(F - X_0))$$

在 original coordinates 中对应 moving cutoff

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)).$$

它 supported in $B_{2R/\lambda}(X_0)$，equal to 1 on $B_{R/\lambda}(X_0)$，随 λ 改变。fixed terminal-limit argument 只控制一个 fixed Q_χ，不能自动控制 moving family $Q_{R,\lambda}$。抽象上，即使每个 Q_λ 都 nonnegative、has terminal limit、并满足 $Q_\lambda' \le 0$，也可能有

$$Q_\lambda(T + \lambda^{-2}s_2) - Q_\lambda(T + \lambda^{-2}s_1) \not\to 0.$$

因此 individual terminal limits for a moving family are insufficient；需要 uniform bridge。

（status: **proven**）standalone literal moving-cutoff lemma 是假的。取 flat symplectic plane $P = \mathbb{R}^2 \times \{0\} \subset \mathbb{C}^2$，stationary flow $F(x,t) = x$。此时 $\cos\alpha = 1$，$H = 0$，所有 curvature/flow-energy terms vanish，且 $\Sigma^\lambda_s = P$。对 nontrivial radial compact cutoff ϕ_R，

$$I(s) = \int_P \phi_R(Y)(-s)^{-1} e^{-|Y|^2/[4(-s)]}\, dA(Y).$$

full uncut integral 恒等于 4π，但 compact-cut integral 不必 constant。写 $a = -s$，change variables $Y = \sqrt{a}\, Z$ 得

$$I(-a) = \int_P \phi_R(\sqrt{a}\, Z) e^{-|Z|^2/4}\, dA(Z).$$

若 $s_1 < s_2 < 0$，则 $a_1 > a_2$。选择 radial nonincreasing cutoff，使 transition annulus 被两种 scales 分开，则

$$I(s_2) > I(s_1).$$

于是

$$G_\lambda(s) = e^{c_1\lambda^{-1}\sqrt{-s}} I(s)$$

给出 positive endpoint defect。这个 plane 不是 full Section 5 counterexample，因为它 noncompact、没有 finite first singular time，且 X_0 不是 singular point。

（status: **proven / conditional**）若 actual endpoint weighted Radon measures 在 s_1, s_2 都 weakly converge 到同一个 nonzero two-dimensional cone measure，则 literal compact-scaled-cutoff endpoint difference 反而通常有 positive limit。原因很简单：对 compact continuous test function

$$Y \mapsto \phi_R(Y)(-s_i)^{-1}e^{-|Y|^2/[4(-s_i)]},$$

weak convergence 直接传递 integrals。对 two-homogeneous cone，radial scaling 又给出 compact cutoff 的 s-dependence。此结论是 conditional obstruction，不是 full counterexample，因为 same-limit conical convergence 不能 circularly 用 later tangent-cone theorem 来提供。

3. 主要障碍

唯一真正的 wall 是 moving cutoff 的 order-one annular flux。把

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0))$$

代入 fixed-cutoff calculation 时，$D\chi_{R,\lambda}$ scales like λ，$D^2\chi_{R,\lambda}$ scales like λ^2。在 parabolic time window $t = T + \lambda^{-2}s$ 中，$\rho_t = \lambda^2\rho_s$，$d\mu_t = \lambda^{-2}d\mu_s^\lambda$，$dt = \lambda^{-2}ds$，这些 powers 正好 cancel，留下

$$\int_{s_1}^{s_2}\int_{\Sigma_s^\lambda} w_\lambda\rho_s\Big[D\phi_R(F_\lambda)\cdot v_\lambda + \Delta_{\Sigma_s^\lambda}(\phi_R\circ F_\lambda) + 2\langle\nabla\log\rho_s, \nabla(\phi_R\circ F_\lambda)\rangle\Big]\,d\mu_s^\lambda ds.$$

该 integral supported in fixed scaled annulus $R < |F_\lambda| < 2R$，没有 small λ factor，也没有 sign。local area bounds 只给 fixed-time compactness；zero velocity/zero curvature heuristics 也不够，因为 stationary plane 上 pure spatial heat-kernel flux already equals the nonzero endpoint defect。later time-independent tangent-cone convergence 不能作为 pre-(5.5) proof，因为它依赖由 (5.5) 推出的 estimates。

4. Approach timeline

stage	question addressed	conclusion established	effect on the approach
fixed terminal quantity	Does one-sided estimate imply a finite terminal limit?	yes, by adding the integrable future error tail	validates the common-limit argument for one fixed cutoff
exact rescaling	What does the fixed original cutoff become after blow-up?	$\chi(F)$ becomes $\chi(X_0 + \lambda^{-1}F_\lambda)$, with a harmless 4π factor	gives a rigorous corrected endpoint formula
literal scaled cutoff	Is $\phi_R(F_\lambda)$ the same fixed-cutoff object?	no; it is $\chi_{R,\lambda}(F)$ in original coordinates	exposes the missing uniform moving-family step
model obstruction	Is fixed compact scaled cutoff harmless?	no; flat stationary plane gives positive endpoint defect	disproves standalone lemma, but not full compact singular-flow case
annular flux audit	Can scaling make moving-cutoff errors small?	no; leading cutoff terms are order-one on a fixed scaled annulus	identifies the exact missing bridge
fixed-time compactness	Can earlier estimates give some compactness?	yes, fixed-time local Radon compactness	insufficient to compare two endpoint measures

5. Current status & next step

literal compact-scaled-cutoff version of equation (5.5) is not proved from the available pre-(5.5) material. The strongest unconditional corrected theorem is the fixed-original-cutoff version above. No full compact first-singular-flow counterexample has been established. The remaining actionable lemma is the following uniform moving-cutoff bridge.

Remaining lemma: uniform moving-cutoff bridge

Let $(\Sigma_t, F(\cdot,t))$ be a compact beta-symplectic critical surface flow smooth for $t < T$, let (X_0, T) be a singular spacetime point, fix $R > 0$, fix $\phi_R \in C_c^\infty(B_{2R}(0))$ with $\phi_R = 1$ on $B_R(0)$, and set

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)).$$

For

$$Q_{R,\lambda}(t) = e^{c_1\sqrt{T-t}} \int_{\Sigma_t} (\cos\alpha)^{-p} \chi_{R,\lambda}(F) \rho_{X_0,T}(F,t)\, d\mu_t,$$

let

$$L_{R,\lambda} = \lim_{t\uparrow T} Q_{R,\lambda}(t),$$

assuming this limit exists from the one-sided estimate for each fixed λ. Prove, noncircularly and uniformly at the parabolic scale, that for every fixed $-1 < s_1 < s_2 < 0$,

$$Q_{R,\lambda}(T + \lambda^{-2} s_i) - L_{R,\lambda} \to 0 \quad (i = 1, 2).$$

Equivalently, prove the annular cutoff-flux terms supported on

$$R < |F_\lambda| < 2R$$

are $o(1)$, or cancel, as $\lambda \to \infty$.